ONE-NAS on CRSP Mid-Cap Panels: Results

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Working draft (results only) — 2026-08-30

Table 1: Net return (%) by trade year, mean over $V1$–$V4$, 10 seeds. ONE-NAS books restarted at each window boundary; the 2022–24 column is one book run over the three-year window.

	2022	2023	2024	2022–24
ONE-NAS (40 isl., sleeves book)	$+6.1_{\pm 0.8}$	$+12.4_{\pm 0.7}$	$+6.4_{\pm 0.5}$	$+27.5_{\pm 1.1}$
ONE-NAS (40 isl., banded book)[a]	$+12.5_{\pm 1.2}$	$+14.6_{\pm 1.3}$	$+3.4_{\pm 0.9}$	$+33.9_{\pm 1.8}$
ONE-NAS (40 isl., Algorithm 1 book)[b]	$+1.3_{\pm 1.6}$	$+12.8_{\pm 1.0}$	$+10.8_{\pm 1.2}$	$+25.0_{\pm 1.8}$
ONE-NAS single champion (8 isl., sleeves)[c]	$+3.1_{\pm 1.8}$	$+1.7_{\pm 1.3}$	$+2.7_{\pm 1.1}$	$+8.5$
Online LSTM	$+1.2_{\pm 1.3}$	$+4.5_{\pm 1.2}$	$+3.8_{\pm 0.7}$	$+11.3_{\pm 1.8}$
Online GRU	$+2.1_{\pm 1.0}$	$+3.1_{\pm 1.9}$	$+5.0_{\pm 0.8}$	$+11.7_{\pm 2.2}$
Periodic LSTM (monthly retrain)	$+1.4_{\pm 1.0}$	$+3.9_{\pm 0.8}$	$+6.9_{\pm 0.6}$	$+14.8_{\pm 1.0}$
Online AR[d]	−14.6	+4.8	+7.9	−3.3
Buy & hold[e]	−10.1	+11.2	+6.8	+7.9

[a] Buy/hold-spread rule (enter rank ≤10, exit past 35); sensitivity row, not an adopted headline.
[b] Conditional long–short book (rebalance only when all top-10 predictions are positive and all bottom-10 negative); on rank-normal predictions the gate fires daily.
[c] Published ONE-NAS prediction rule; seeds 42–44 (n=3).
[d] Deterministic; one run per panel. All baselines trade the sleeves book under identical costs.
[e] Equal-weight, prior-study benchmark convention; 2022–24 column is the sum of yearly cells.

Table 2: Prediction rule, within-run paired: single global-best genome vs. island-champion rank-mean ensemble from the same runs (sleeves book). Net % / Sharpe; Δ = ensemble − single, paired t over seeds.

Fleet	Window	Single	Ensemble	Δnet (t)	ΔSharpe (t)
40 isl. (n=10)	2022–24	+4.5 / 0.21	+27.5 / 0.87	+23.0 (18.2)	+0.66 (14.7)
40 isl. (n=10)	2020–24	+2.7 / 0.08	+45.6 / 0.68	+42.9 (14.6)	+0.60 (13.4)
8 isl. (n=10)	2022–24	+8.3 / 0.33	+20.5 / 0.70	+12.2 (6.7)	+0.37 (5.8)
8 isl. (n=10)	2020–24	+12.0 / 0.21	+32.1 / 0.52	+20.1 (5.5)	+0.30 (5.3)
8 isl. (n=5)[a]	2016–19	+16.3 / 0.56	+26.0 / 0.92	+9.7 (5.2)	+0.36 (4.5)
40 isl. (n=5)[a]	2016–19	+8.9 / 0.30	+36.7 / 1.22	+27.8 (7.5)	+0.92 (6.6)

[a] Replication on the 2016–2019 tuning-span fleets.

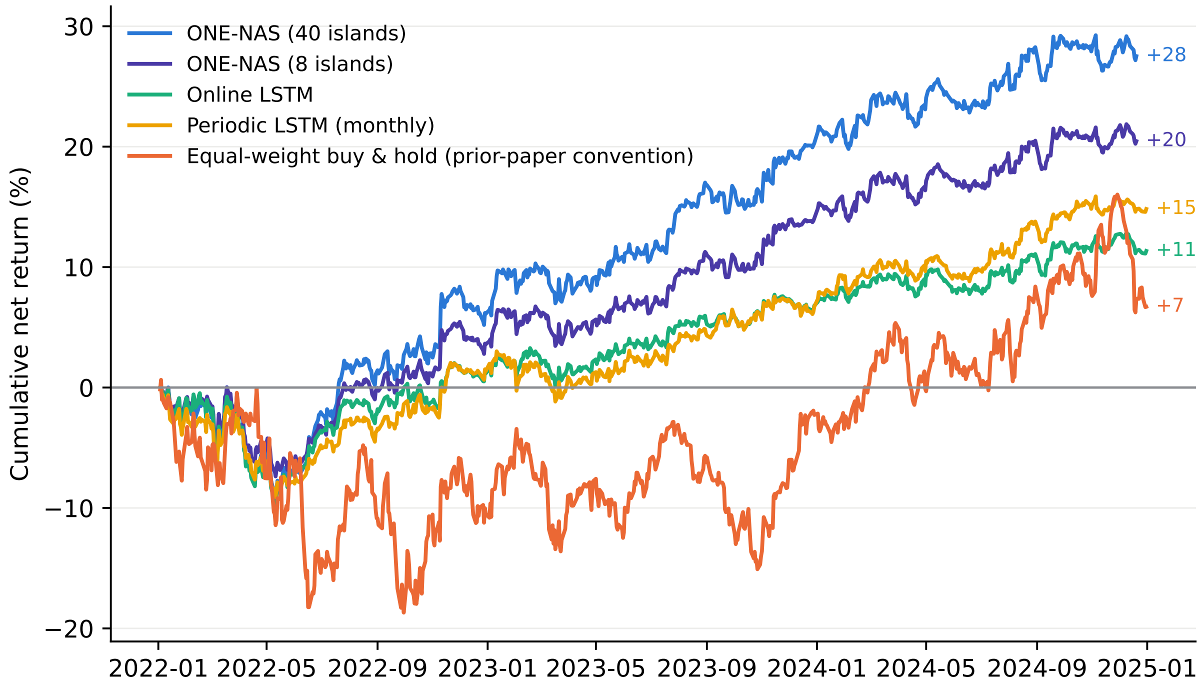

Figure 1: Cumulative net return, 2022–2024 (books restarted at window start). Buy-and-hold follows the prior paper's benchmark convention.

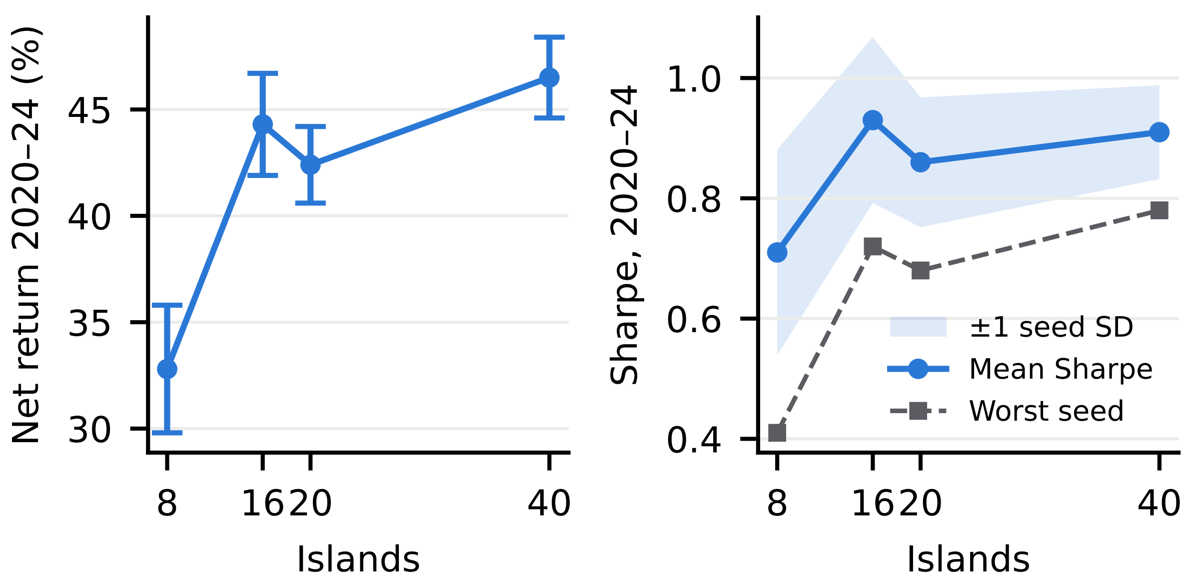

Figure 2: Performance vs. island count (2020–2024, registered sleeves book; 10 seeds per width, 6 at 16 islands). Left: pooled net return (mean ± SE). Right: Sharpe — mean with ±1 seed-SD band, and the worst seed; width narrows the seed distribution (SD 0.171 → 0.078, worst seed 0.41 → 0.78) more than it moves the mean.

Table 3: Factor-adjusted alphas: pooled daily book returns regressed on Fama–French three factors plus momentum, 2020–2024, Newey–West lag 10.

Arm	α (bps/day)	α (%/yr)	NW t	Mkt. beta
ONE-NAS (40 islands)	+3.15	+7.9	+2.14	+0.120
ONE-NAS (8 islands)	+2.17	+5.5	+1.76	+0.096
Periodic LSTM (monthly)	+2.15	+5.4	+1.78	+0.079
Online LSTM	+2.08	+5.2	+1.59	+0.130
Online GRU	+1.52	+3.8	+1.35	+0.119
Online AR	−1.35	−3.4	−1.05	+0.007

Table 4: Ensemble diversity mechanism: measured member rank IC, pairwise member prediction correlation ρ, measured ensemble IC, and the equicorrelated averaging prediction $m\sqrt{N/(1+(N-1)\rho)}$.

Fleet	N	ρ	Member IC	Ensemble IC	Predicted
8 isl., 2020–24	8	0.216	+0.0064	+0.0110	+0.0114
40 isl., 2020–24	40	0.172	+0.0058	+0.0128	+0.0132
8 isl., 2016–19	8	0.190	+0.0071	+0.0132	+0.0132
40 isl., 2016–19	40	0.187	+0.0078	+0.0166	+0.0171

Table 5: Ensembling gain (ensemble − single champion, within-run, net % on 2016–2019) inside every trained hyperparameter configuration; positive in 16/16 cells.

Configuration	Δnet	Configuration	Δnet
C0 (registered primary)	+3.9	NTS 600	+12.8
Select: IC-gated	+0.5	Rounds 2	+11.0
Select: IC	+2.8	PER α 0.4	+4.0
Target: 5-day CS	+4.5	PER α 0.8	+11.7
5-day CS + IC-gated	+1.8	PER $\lambda\times3$	+20.6
Elite 5 / gen 10	+9.6	BP epochs 20	+12.5
Elite 5 / gen 15	+11.1	Repop. 25	+7.0
Repop. 100	+9.6	PER off (uniform)	+9.3

Mean +8.3; n=6 runs per cell (2 panels × 3 seeds).

Table 6: Online continuity: pooled net (%) on 2022–2024, frozen 8-island configuration.

Population evolving online since 2020-01	+26.5
Cold start at 2022-01	+12.3

Table 7: ONE-NAS hyperparameter settings (selected on 2016–2019 validation data, disjoint from the evaluation span). Replay sampling ablated: PER vs. uniform shows no advantage on the tuning objective (ΔIC -0.0024, $t{=}-1.9$).

Hyperparameter	Value
Islands	40 (headline); 8 (registered primary)
Elite population per island	8
Generated genomes per island per generation	5
Repopulation frequency	every 50 generations
Selection metric	validation MSE
Prediction rule	rank-mean ensemble of island champions
Target	1-day cross-sectional rank-normal return
Sequence length / window step / validation sets	40 / 5 / 5
Training subsequences per genome	2000
Backpropagation epochs per genome	10
Replay (PER) α, λ, ϵ	0.6, 0.007, 10^{-8}
Rounds per generation	1
Node-type library	simple, UGRNN, MGU, GRU, Δ-RNN, LSTM
Mutation / intra- / inter-island crossover	0.7 / 0.2 / 0.1 (framework defaults)
Warm-up generations	50